Decomposition-Ensemble Approach for Realized Volatility Prediction

Published: 18 June 2025

(2025) Cite this article

(IMFs) and residuals, effectively capturing the intricate patterns inherent in the data. Sample entropy and t-tests are utilized to classify these components into high-frequency and low-frequency segments. The low-frequency components, representing long-term trends, are directly fed into various Deep Neural Network (DNN) architectures, including Simple Recurrent Neural Network (SRNN), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Bidirectional GRU (BiGRU), and Bidirectional LSTM (BiLSTM). High-frequency components, capturing short-term fluctuations, are modeled using Generalized Autoregressive Conditional Heteroskedasticity (GARCH) to extract conditional volatility, which is subsequently inputted into the DNN models. The final realized volatility prediction is obtained by linearly superimposing the outputs from all components, forming a comprehensive ensemble model. The proposed hybrid models are benchmarked against traditional standalone DNN models using evaluation metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and R-squared (R^2). Empirical results demonstrate that the hybrid VMD-GARCH-DNN models

Explore related subjects

Discover the latest articles, books and news in related subjects, suggested using machine learning.

- Financial Econometrics
- Machine Learning
- Neural decoding
- Quantitative Finance
- Statistical Finance
- Structure Prediction
- Financial Time Series Prediction Using Machine Learning Techniques

Data Availability

Data available on request and the codes it will be available to public once the manuscript has

Becha, M., Dridi, O., Riabi, O., & Benmessaoud, Y. (2020). Use of machine learning techniques in financial forecasting. In *2020 International Multi-Conference on:"Organization of Knowledge and Advanced Technologies"(OCTA)*, pages 1–6. IEEE

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, *31*(3), 307–327.

Article Google Scholar

Bollerslev, T., Chou, R. Y., & Kroner, K. F. (1992). Arch modeling in finance: A review of the theory and empirical evidence. *Journal of Econometrics*, *52*(1–2), 5–59.

Article Google Scholar

Article Google Scholar

Christensen, K., Siggaard, M., & Veliyev, B. (2023). A machine learning approach to volatility forecasting. *J. Financ. Economet.*, *21*(5), 1680–1727.

Article Google Scholar

Chung, J., Gulcehre, C., Cho, K., & Bengio, Y. (2014). Empirical evaluation of gated recurrent neural networks on sequence modeling. arXiv:1412.3555

Corsi, F., Mittnik, S., Pigorsch, C., & Pigorsch, U. (2008). The volatility of realized volatility. *Economet. Rev.*, *27*(1–3), 46–78.

Di Persio, L., Garbelli, M., Mottaghi, F., & Wallbaum, K. (2023). Volatility forecasting with hybrid neural networks methods for risk parity investment strategies. *Expert Syst. Appl.*, *229*, Article 120418.

Article Google Scholar

Di Persio, L., Garbelli, M., & Wallbaum, K. (2021). Forward-looking volatility estimation for risk-managed investment strategies during the covid-19 crisis. *Risks*, *9*(2), 33.

Article Google Scholar

Dragomiretskiy, K., & Zosso, D. (2013). Variational mode decomposition. *IEEE Trans. Signal Process.*, *62*(3), 531–544.

Hendricks, D. (1996). Evaluation of value-at-risk models using historical data. *Economic Policy Review*, *2*(1)

Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Comput.*, *9*(8), 1735–1780.

Article Google Scholar

Hu, Y., Ni, J., & Wen, L. (2020). A hybrid deep learning approach by integrating lstm-ann networks with garch model for copper price volatility prediction. *Physica A*, *557*, Article 124907.

Article Google Scholar

Kambouroudis, D. S., McMillan, D. G., & Tsakou, K. (2016). Forecasting stock return volatility: A comparison of garch, implied volatility, and realized volatility models. *J. Futur. Mark.*, *36*(12), 1127–1163.

Article Google Scholar

Kim, T. K. (2015). T test as a parametric statistic. *Korean J. Anesthesiol.*, *68*(6), 540–546.

Article Google Scholar

Kim, H. Y., & Won, C. H. (2018). Forecasting the volatility of stock price index: A hybrid model integrating lstm with multiple garch-type models. *Expert Syst. Appl.*, *103*, 25–37.

Article Google Scholar

Li, H., Zhou, D., Hu, J., Li, J., Su, M., & Guo, L. (2023). Forecasting the realized volatility of energy stock market: A multimodel comparison. *The North American Journal of Economics and Finance*, *66*, Article 101895.

Article Google Scholar

Lu, X., Ma, F., Xu, J., & Zhang, Z. (2022). Oil futures volatility predictability: New evidence based on machine learning models. *Int. Rev. Financ. Anal.*, *83*, Article 102299.

Article Google Scholar

Medsker, L. R., & Jain, L. (2001). *Recurrent neural networks. Design and Applications*, *5*(64–67), 2.

Richman, J. S., Lake, D. E., & Moorman, J. R. (2004). Sample entropy. In *Methods in enzymology*, volume 384, pages 172–184. Elsevier

Ryll, L., & Seidens, S. (2019). Evaluating the performance of machine learning algorithms in financial market forecasting: A comprehensive survey. arXiv:1906.07786

Schuster, M., & Paliwal, K. K. (1997). Bidirectional recurrent neural networks. *IEEE Trans. Signal Process.*, *45*(11), 2673–2681.

Article Google Scholar

Seo, M., & Kim, G. (2020). Hybrid forecasting models based on the neural networks for the volatility of bitcoin. *Appl. Sci.*, *10*(14), 4768.

Wang, D., Yue, C., Wei, S., & Lv, J. (2017). Performance analysis of four decomposition-ensemble models for one-day-ahead agricultural commodity futures price forecasting. *Algorithms*, *10*(3), 108.

Article Google Scholar

Wei, Q., Wu, B., Li, X., Guo, X., Teng, Y., Gong, Q., & Wang, S. (2024). Motion interval prediction of a sea satellite launch platform based on vmd-qr-gru. *Ocean Eng.*, *312*, Article 119005.

Article Google Scholar

Wen, F., Yang, X., Gong, X., & Lai, K. K. (2017). Multi-scale volatility feature analysis and prediction of gold price. *International Journal of Information Technology & Decision Making*, Zhang, Y., Peng, Y., & Song, Y. (2024). Realized volatility forecasting for stocks and futures indices with rolling ceemdan and machine learning models. *Computational Economics*, pages 1–54

Zhang, C., Zhang, Y., Cucuringu, M., & Qian, Z. (2024). Volatility forecasting with machine learning and intraday commonality. *J. Financ. Economet.*, *22*(2), 492–530.

Article Google Scholar

Zhou, F., Huang, Z., & Zhang, C. (2022). Carbon price forecasting based on ceemdan and lstm. *Appl. Energy*, *311*, Article 118601.

Article Google Scholar

Authors and Affiliations

School of Management, University of Science and Technology of China , Jinzhai Road, Hefei, 230026, People's Republic of China

John Kamwele Mutinda & Li Yong

Contributions

John Kamwele Mutinda: Writing -Original draft, Writing-Review, Editing, Software, Methodology, Data curation and Conceptualization **Li Yong**: Writing-Review, Editing and Supervision

Corresponding author

Correspondence to John Kamwele Mutinda.

self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Reprints and permissions

About this article

Cite this article

Mutinda, J.K., Yong, L. Decomposition-Ensemble Approach for Realized Volatility Prediction. *Comput Econ* (2025). https://doi.org/10.1007/s10614-025-11020-0

Accepted	Published	Version of record
04 June 2025	18 June 2025	18 June 2025